

and the entire manufacturing supply chain is connected via a digital thread. This isn't a vision for the future; it's the reality many manufacturers are embracing today.

Essentially, Industry 4.0 is powered by IT-driven technologies like the Internet of Things (IoT), AI, cloud and edge computing, and big data analytics, with the emerging interventions from Generative AI.

However, the importance of Industry 4.0 goes beyond efficiency. These advancements are helping manufacturers rethink their environmental impact right from the product conceptualisation and design stage. For instance, IoT sensors enable precise energy and water usage monitoring, ensuring that factories run as sustainably as possible.

Similarly, AI-driven tools can optimise resource allocation, reducing waste and promoting smarter practices across the board.

This intersection of technology and manufacturing isn't just a leap forward in capability; it's also an opportunity to embed sustainability into the very fabric of how products are made, consumed, and sustained.

It's often said that what gets measured gets managed. IT allows manufacturers to measure and manage their operations in ways that weren't possible before.

The insights generated through IT enable businesses to achieve a balance between innovation and responsibility—a balance that has never been more critical.

For example, AI-powered design systems can simulate the eventual sustainability impact of a product concept and optimise design features; robotics-driven advanced manufacturing systems can convert product designs to manufactured goods efficiently, while IoT-powered self-healing equipment can predict failures before they happen. Blockchain technology adds another layer of accountability, offering transpar-

ency and traceability across supply chains.

These technologies form the backbone of a manufacturing ecosystem that is smart and kinder to the planet. In a world increasingly defined by finite resources, IT serves as the bridge between operational excellence and environmental stewardship.

India's Quest

India's journey towards becoming a global manufacturing hub is accelerating, powered by Industry 4.0. According to a Nasscom report on India's Industry 4.0 adoption, digital technologies are projected to account for 40% of total manufacturing expenditures by 2025 compared to 20% in 2021. As the Indian manufacturing sector continues to adopt these technologies rapidly, the government too has taken significant strides to promote manufacturing, technology, and sustainable practice adoption.

Initiatives like 'Make in India' and 'Digital India' have created a favourable environment for businesses to invest and innovate. The production-linked incentive (PLI) scheme has been instrumental in attracting investments in key manufacturing sectors. Put these together, and we enhance India's unique position as

the melting pot of pivotal priority areas.

At the same time, a skilled workforce that understands and leverages these tools is essential for sustainable growth. The government's focus on skill development and technological upgradation has empowered the workforce to embrace Industry 4.0.

Blueprint for the Future

The intersection of IT, manufacturing operations technology (OT), and sustainability in Industry 4.0 represents a once-in-a-generation opportunity. With the interventions of Gen AI and a focus on building large language models (LLMs) enabling smart digital manufacturing, manufacturing firms have significant opportunities to improve overall product design and throughput and predict manufacturing line performance. This convergence of OT and IT systems is building on the proven technological foundation and maturity of IT to deliver unparalleled insights across enterprises. This has paved the path for truly differentiated value, transforming IT into a critical enabler for resilient, expansive, adaptive, and integrated enterprises.

India's strong IT infrastructure, policy framework, and talent pool provide the foundation for leading this transformation. By continuing to adopt sustainable practices, invest in innovation, and empower its workforce, it can truly emerge as a global manufacturing powerhouse that prioritises both growth and the planet.

The IT blueprint for Industry 4.0 is here and clear—smarter systems, responsible choices, and a sustainable future for all. **BT**

Views are personal



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